

1st Outline

Intro

In this section, I will introduce how the Pomona-Pitzer Women's Water Polo Team collects data from tagged Hudl game film. The team already has a detailed shot-analysis workflow, however, the process takes a long time, requiring substantial manual work. My project will look at if an altered workflow can help in decreasing the amount of time it takes to collect detailed shot-data and analyses, while still preserving the amount of details and information the coaches want and need. My goal is to develop a system that uses the tagged Hudl game film to create a usable shot dataset.

Related Analyses

This section will talk about related research already done for sports analyses, including water polo and different sports, game film annotation, and sports analytics. I will talk about studies that have looked at characteristics of shots like shot location, shot type, and shooting success, and research on expected goal models in other sports. Previous and related works will help me with the variables I choose to use and guide my project for game film data collection and what variables or shot characteristics to use.

The Data and Study Design

In this section, I will discuss the data being used in this project. I am going to use game film from the Pomona-Pitzer Women's Water Polo 2025-2026 season. I am only going to use games where the PPWWP team played teams that were ranked in the top 25, as well as Claremont-Mudd-Scripps Women's water polo team, PPWWP's rivals. I will explain what information is already in Hudl and what information is not there that needs to be added. I will define my data and unit of observation, which is one shot, and all other relevant variables.

Development of Shiny App. and Data-Curation Pipeline

This is one of my main sections, where I describe the workflow pipeline I am actually building. I will describe the Shiny application and what I am developing to make the existing shot-analysis hopefully more efficient. Ideally, the program takes the Hudl Sportscube exported Excel sheet that has timestamps of the shots for one game, and allows the user to move through the timestamps while selecting the standardized attributes attached to the shot. The program will save the selections and organize the shots in a dataset.

Analysis

After I make the dataset, I will analyse which shot characteristics are associated with successful shots (goals). My primary outcome variable will be on whether a shot was a goal. I plan to run a regression on the outcome variable and shot attribute variables. Depending on the timeline of this project, I may aim to make more analyses or a visual dashboard for what variables are attributable to successful shots.

Results

This section will present the results from the analysis. I will share the overall shooting patterns and regression. I will focus on interpreting the results in a way that is most useful for my coaches to know. I may even add my own observations, being a water polo player myself, as well as my coach's notes.

Ethics and Limitations

I will discuss the ethical limitations and considerations of using this data. While I do not have too many limitations, as most of this information is public, it is important to consider the ethical limitations of player privacy and human errors that happen in the initial tagging of the shots. I will also discuss reproducibility. If this workflow is more efficient than documenting the standardized variables and definitions and documenting the workflow will be important for it being repeated in the future.

Conclusion/Discussion

This final section will summarize the project and findings. I will discuss if the project was ultimately more efficient and if not, what would make it more efficient. I will discuss what the data reveals about the PPWWP team's shots and shooting pattern and how the system could be used in the future. I could discuss expanding the data to multiple seasons, other teams, adding more shooting characteristics, or contextual variables.